Quantum Field Theory

Lecture Notes

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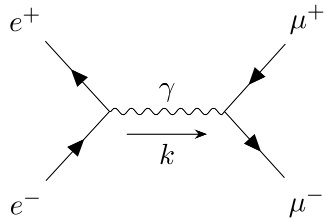

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Lecture 01: Quantum SHO

People do 'weird' stuffs for earning,
You can do the same for learning!

Since the foundational aspect of QFT (and many other topics in Physics) is the Harmonic Oscillator, let us discuss the quantum Harmonic oscillator for introduction. For that, note that the classical Hamiltonian for the harmonic oscillator is given by:

$$H = \frac{p^2}{2m} + \frac{1}{2}m\omega^2 x^2$$

We do not like the things like m and ω which prevent us from seeing things clearly 😅. Hence, we invoke the holy action of making things dimensionless. For that, we define:

$$\begin{aligned} X &= \sqrt{\frac{m\omega}{\hbar}}x &\longrightarrow&& x^2 &= \frac{\hbar}{m\omega}X^2 \\ P &= \frac{1}{\sqrt{m\omega\hbar}}p &\longrightarrow&& p^2 &= m\omega\hbar P^2 \end{aligned}$$

Substituting these in the Hamiltonian, we have:

$$H = \frac{\hbar\omega}{2}(X^2 + P^2)$$

Note that, we are still within the classical domain. Now, let us elevate x and p to operators and we define:

$$[x, p] = i\hbar\, \mathbb{1} \implies [X, P] = \sqrt{\frac{\cancel{m\omega}}{\hbar}}\frac{i\hbar\, \mathbb{1}}{\sqrt{\cancel{m\omega\hbar}}} = i$$

Introducing the commutator bracket brings us to the quantum world. Now, we invoke our very own ladder operators:

$$\begin{aligned} \hat{a} &= \frac{1}{\sqrt{2}}(\hat{X} + i\hat{P}) &&: \text{annihilation operator} \\ \hat{a}^\dagger &= \frac{1}{\sqrt{2}}(\hat{X} - i\hat{P}) &&: \text{creation operator} \end{aligned}$$

Then we will have [1]:

$$\left[a, a^\dagger\right] = \frac{1}{2}[X + iP, X - iP] = \frac{-i}{2}([X, P] - [P, X]) = -i \times i = 1$$

Also, note that:

$$a^\dagger a = \frac{1}{2}(X^2 + P^2 + i\underbrace{(XP - PX)}_{i}) = \frac{1}{2}(X^2 + P^2) - \frac{1}{2} \implies H = \frac{\hbar\omega}{2}(a^\dagger a + \frac{1}{2})$$

Let us consider the *complete set of commuting observables* (CSCO) for this problem. Evidently, the set $\{H\}$ itself satisfies the condition since the eigenvalues are all non-degenerate (hence, we can label each state with only one index). To understand why, let us consider the action of the the annihilation operator on a (normalised) state $|\psi\rangle$. For that we note the following:

$$\begin{aligned} \left[a^\dagger a, a\right] &= -a \implies [H, a] = -\hbar\omega\ a \\ \left[a^\dagger a, a\right] &= a^\dagger \implies \left[H, a^\dagger\right] = \hbar\omega\ a^\dagger \end{aligned}$$

[1] henceforth, forsaking the hat symbol and identity operator $\mathbb{1}$, since they cause nothing but trouble, when the context is clear

Now, we have:

$$Ha\left|\psi\right\rangle - aH\left|\psi\right\rangle = [H,a]\left|\psi\right\rangle = -\hbar\omega a\left|\psi\right\rangle \implies H(a\left|\psi\right\rangle) = (E-\hbar\omega)(a\left|\psi\right\rangle)$$

Thus, if $\left|\psi\right\rangle$ has an energy eigenvalue E, then $a\left|\psi\right\rangle$ will have an energy eigenvalue $E-\hbar\omega$. Thus, starting from any eenrgy state, we can change to another state with energy reduced by one unit of $\hbar\omega$, using the annihilation operator. Similarly, we will have:

$$H(a^\dagger\left|\psi\right\rangle) = (E+\hbar\omega)(a^\dagger\left|\psi\right\rangle)$$

Let us denote the states $a^\dagger\left|\psi\right\rangle$ and $a\left|\psi\right\rangle$ by $\left|\psi_+\right\rangle$ and $\left|\psi_-\right\rangle$ respectively. Then, we will have

$$\left\langle\psi_-|\psi_-\right\rangle = \left\langle\psi\right| a^\dagger a\left|\psi\right\rangle = \left\langle\psi\right|\left(\frac{H}{\hbar\omega}-\frac{1}{2}\right)\left|\psi\right\rangle$$

Now, since $\left|\psi_-\right\rangle$ is a valid vector of the Hilbert space, its norm must be non-negative and finite. Hence, we obtain the condition:

$$0 \le \frac{E}{\hbar\omega} - \frac{1}{2} < \infty \implies \frac{\hbar\omega}{2} \le E$$

Hence, we get a lower bound on the energy eigenvalue, that is, there exists a state $\left|\psi_{\text{min}}\right\rangle$ such that $H\left|\psi_{\text{min}}\right\rangle = E_{min}\left|\psi_{\text{min}}\right\rangle$ where $E_{min} = \frac{\hbar\omega}{2}$. Then, starting from this state, if we apply the creation operator, we will get successively increasing energies (and hence the system is non-degenerate). We thus obtain:

$$E_n = \left(n+\frac{1}{2}\right)\hbar\omega \qquad n = 0, 1, \ldots$$

Lecture 02: Dimensional Deep Dive

For the quantum harmonic oscillator, we had seen that how making things dimensionless eases calculations a bit. We consider three fundamental constants, each occupying their own special place in their field of action.

$$\begin{aligned} c &\equiv [LT^{-1}] && : \text{ relativity} \\ \hbar &\equiv [L^2MT^{-1}] && : \text{ quantum} \\ G &\equiv [L^3M^{-1}T^{-2}] && : \text{ gravity} \end{aligned}$$

Lengths are tractable for us, since we can see the 'length', same goes for mass, atleast we can 'feel' it. However, time is an *enigma. We can't hold two ends of time at the same time* unlike holding two ends of a rod to measure its length. The absolute truth is: *Time passes!*

Note that, in physics, we are mostly concerned with equations like $E = mc^2$ and $E = \hbar\omega$. To that extend, we define the natural units:

$$\begin{aligned} c &= 1 \\ \hbar &= 1 \end{aligned}$$

We want these quantitites to be numerically equal to 1 and dimensionless. Note that, we had also done this kind of things before. When writing Newton's law, we had said that

$$F \propto m, F \propto a \implies F \propto ma \implies F = kma \qquad \text{for some } k$$

Now, we chose unit and dimension of force in a way such that $k = 1$ (dimensionless and unit value) which gave us the celebrated law.

Note that:

- Making c dimensionless:
$$[LT^{-1}] \equiv 0 \implies [L] = [T]$$
- Making $\hbar$ dimensionless:
$$[L^2MT^{-1}] = [L^2ML^{-1}] = [LM] \equiv 0 \implies [L] = [M^{-1}]$$

Hence, we see that length and time are equivalent while mass and length have inverse relation. In this natural units, we have that energy is equivalent to mass and any other unit can be represented in terms of mass. Thus, by our convention, we choose mass or energy as the only important dimension.

Note that
$$\hbar c \equiv Jm = 1 \qquad \text{(in natural units)}$$
From this, we can say heuristically that increasing length scale is decreases the energy (mass) scale. We mainly use the length scale in context of *de Broglie wavelength*.

This is perhaps not the only way to define natural units. In cosmology, G plays a more important role and hence it is better to set $G = 1$, leaving aside $\hbar$. Using this *natural units*, we find:
$$[L] = [M]$$
$$[L] = [T]$$

Here, we see that length and mass scale are directly related. Well, in this regard, we treat the length scale to be that of the Schwarzschild radius of a blackhole [1] which intuitively grows with mass. Since the above two natural units are widely distinct, there is as such no problem, however, in the unfortunate case where we have to consider both de Broglie wavelength and the Schwarzschild radius (that is, in the infamous domain of quantum gravity 😨), one needs to be very careful.

Theorem 1 ($\pi-$Theorem):
Let $q_1, \ldots, q_n$ be n variables which are physically relevant to a problem and which are related by an expression, that is,
$$F(q_1, \ldots, q_n) = 0 \iff q_i = \widetilde{F}(q_1, \ldots \hat{q}_i, \ldots q_n)$$
If k is the number of fundamental dimensions required to describe the n variables, then we can group these in $(n-k)$ groups of dimensionless variables $\Pi_1, \ldots, \Pi_{n-k}$ such that for some f, we have:
$$f(\Pi_1, \ldots, \Pi_{n-k}) = 0 \iff \Pi_i = \widetilde{f}(\Pi_1, \ldots \hat{\Pi}_i, \ldots \Pi_{n-k})$$
.

The theorem seems a bit vague (and pointless too). Let us take a physical example. Consider a spherical ball in a viscous fluid. The variable in the problem are:
$$\begin{aligned}
\text{Drag force:} \quad & q_1 \to F \quad [MLT^{-2}] \\
\text{Sphere diameter:} \quad & q_2 \to d \quad [L] \\
\text{Fluid density:} \quad & q_3 \to \rho \quad [ML^{-3}] \\
\text{Fluid velocity:} \quad & q_4 \to v \quad [LT^{-1}] \\
\text{Fluid viscosity:} \quad & q_5 \to \eta \quad [ML^{-1}T^{-1}]
\end{aligned}$$

[1] This crap is the radius of an object such that if the body is squeezed to a radius lesser than the Schwarzschild radius, the gravitational attraction between the constituents of the body causes its irreversible collapse, turning it to a black hole ⚫

So, there are 5 such parameters and only three units viz. M, L, T are needed to describe them. Hence we will have two Π groups. It is a good thing to choose the repeating variables (variables which will be in both groups) that relate to mass, geometry and the kinematics of the problem. Also, note that since the Π groups are dimensionless, we can take the non-repeating variable's power to be 1. Hence, in this problem we choose them to be ρ, d, v. Thus, we will have:

$$\Pi_1 = \rho^{a_1} d^{a_2} v^{a_3} F \equiv [ML^{-3}]^{a_1}[L]^{a_2}[LT^{-1}]^{a_3}[MLT^{-2}] = [M^{a_1+1}L^{-3a_1+a_2+a_3+1}T^{-a_3-2}]$$
$$\Pi_2 = \rho^{b_1} d^{b_2} v^{b_3} \eta \equiv [ML^{-3}]^{b_1}[L]^{b_2}[LT^{-1}]^{b_3}[ML^{-1}T^{-1}] = [M^{b_1+1}L^{-3b_1+b_2+b_3-1}T^{-b_3-1}]$$

Hence we obtain two sets of equations:

$$\begin{aligned}
a_1 + 1 = 0 &\implies a_1 = -1 \\
-a_3 - 2 = 0 &\implies a_3 = -2 \\
-3a_1 + a_2 + a_3 + 1 = 0 &\implies 3 + a_2 - 2 + 1 = 0 \implies a_2 = -2 \\
b_1 &= -1 \\
b_3 &= -1 \\
-3b_1 + b_2 + b_3 - 1 = 0 &\implies 3 + b_2 - 2 = 0 \implies b_2 = -1
\end{aligned}$$

Then we obtain the Π groups as:

$$\Pi_1 = \frac{F}{\rho d^2 v^2} \qquad \Pi_2 = \frac{\eta}{\rho d v} = \frac{1}{\frac{\rho d v}{\eta}}$$

We identity $\frac{\rho d v}{\eta}$ to be the *Reynold's numer* $\mathcal{R}$. Then we can say, for some ϕ,

$$\frac{F}{\rho d^2 v^2} = \phi(\mathcal{R})$$

To some extraterrestrial being, the physical laws will (hopefully) be valid, however, they might not understand the human-made units (like metres and seconds) in which we measure these quantitites.

We need something natural, based on Nature and hence we used the natural units. For this purpose, we will use things like $c, \hbar, G, k_b \ldots$ and we set all of them to 1. Doing this will lead to change in all scales. For that, we define the Planck units, made of fundamental constants of Nature:

- **Planck mass**: $E_p = \sqrt{\frac{\hbar c}{G}}$
- **Planck length**: $l_p = \sqrt{\frac{\hbar G}{c^3}}$
- **Planck time**: $t_p = \sqrt{\frac{\hbar c}{G^5}}$

Let us now analyse the physical regimes in which the fundamental constants become important. For that, we will use a tuple $(G, \frac{1}{c}, \hbar)$ (note that all the elements of the tuple are written in terms of very small quantitites of the SI scale).

$$\begin{aligned}
(0,0,0) &\to \text{ Classical mechanics} \\
(G,0,0) &\to \text{ Newtonian gravity} \\
\left(0,\tfrac{1}{c},0\right) &\to \text{ Special relativity} \\
(0,0,\hbar) &\to \text{ Basic quantum mechanics} \\
\left(G,\tfrac{1}{c},0\right) &\to \text{ General relativity} \\
\left(0,\tfrac{1}{c},\hbar\right) &\to \text{ QFT and relativistic QM} \\
(G,0,\hbar) &\to \text{ Non-relativistic gravity} \\
\left(G,\tfrac{1}{c},\hbar\right) &\to \text{ Quantum gravity}
\end{aligned}$$

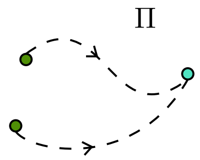
Lecture 03: From Classical to Quantum

Inspecting the transition from a classical to quantum description is necessary for understanding QFT, which is just a *sophisticated mechanics*.

We know that the position $\mathbf{q} \equiv \{q_1, \ldots, q_n\}$ and momentum $\mathbf{p} \equiv \{p_1, \ldots, p_n\}$ uniquely define the state of a *classical* particle.

$$(\mathbf{q}, \mathbf{p}) \in \Pi \text{ (}2n\text{ dimensional phase space)} \qquad q_i, p_i \in \mathbb{R}$$

The Hamiltonian $H(\mathbf{q}, \mathbf{p})$ is just a way of performing time evolution on the system. Using the Hamiltonian, we can 'fibrate' the phase space, that is, starting from one point, we can go to some other point. Also, note that, changing the parameters in the Hamiltonian (eg. ω in SHO Hamiltonian), we change the fibration patterns(eg. *Lissajou's* figures).



In the quantum world, the phase space changes to the Hilbert space $\mathcal{H}$ while a point in the phase space (q, p) changes to a vector $|\psi\rangle$ in the Hilbert space.

Also, these real variables become Hermitian (self-adjoint) operators [1]and the classical Poisson bracket now transforms to the commutator.

$$\begin{aligned} \text{CM:} \qquad & \{q, p\} = 1 \\ \text{QM:} \qquad & [\hat{q}, \hat{p}] = i \end{aligned}$$

3.1 Time Evolution

From classical Hamilton's equations of motion, we have:

$$\begin{aligned} \dot{q}_i &= \{q_i, H\} = \frac{\partial H}{\partial p_i} \\ \dot{p}_i &= \{p_i, H\} = -\frac{\partial H}{\partial q_i} \end{aligned}$$

If $\mathbf{z} = \begin{pmatrix} \mathbf{p} \\ \mathbf{q} \end{pmatrix}$ is a $2n$ dimensional vector, then $\dot{\mathbf{z}} = \{\mathbf{z}, H\}$. Now, moving to quantum mechanics, we know the celebrated Schrödinger equation, which specifies the time evolution of a state vector $|\psi(t)\rangle \in \mathcal{H}$:

$$i\frac{\partial |\psi(t)\rangle}{\partial t} = H |\psi(t)\rangle$$

[1] A self-adjoint operator $\mathcal{O}$ is such that $\mathcal{O} = \mathcal{O}^\dagger$ in all respect, that is, $\mathcal{O}$ and $\mathcal{O}^\dagger$ have the same domain and action. In general, $\mathbb{D}(\mathcal{O}) \subseteq \mathbb{D}(\mathcal{O}^\dagger)$ where $\mathbb{D}(\cdot)$ represents the domain of some operator. If the domains are not equal but action is same on a restricted domain (mainly occurs in infinite-dimensional spaces), then those operators are not self-adjoint but called *symmetric/Hermitian* (in some places Hermiticity also requires a symmetric operator to be bounded).

Thus, in both classical and quantum case, the time derivative of a quantity is equal to some action of the Hamiltonian on that quantity (classical $\to$ Poisson bracket, quantum $\to$ Multiplication with Hamiltonian)[1].

The Hamiltonian in the position basis becomes:

$$H = \frac{-\hbar^2}{2m}\boldsymbol{\nabla}^2 + V(\mathbf{x})$$

Thus, the Schrödinger equation now acts on the wavefunction $\psi(\mathbf{x}, t) = \langle \mathbf{x}|\psi\rangle$:

$$i\frac{\partial\psi(\mathbf{x}, t)}{\partial t} = -\frac{1}{2m}\boldsymbol{\nabla}^2\psi(\mathbf{x}, t) + V(\mathbf{x})\psi(x, t)$$

And the energy and momentum operators become $E \to i\frac{\partial}{\partial t}$ and $p \to -i\boldsymbol{\nabla}$.

3.1.1 Free Particle

The dispersion relation for a non-relativistic free particle is:

$$E = \frac{\mathbf{p}^2}{2m}$$

The typical solution can be written as plane-waves, $e^{i(\mathbf{k}\cdot\mathbf{x}-\omega t)} = e^{i(\mathbf{p}\cdot\mathbf{x}-Et)} \equiv \exp(-ip^\mu x_\mu)$ where we have used the repeated summation notation and taken the metric $\eta_{\mu\nu} = (+, -, -, -)$.

For a relativistic free particle, the dispersion relation becomes:

$$E^2 = m^2 + p^2$$

Instead of the wavefunction $\psi(\mathbf{x}, t)$, we will use $\phi(\mathbf{x}, t)$, since the resulting equation actually acts on a scalar-field and ϕ is common notation for a scalar-field. Substituting the energy and momentum operators here, we get:

$$\begin{aligned}
i^2\frac{\partial^2\phi(\mathbf{x}, t)}{\partial t^2} &= (-i)^2\boldsymbol{\nabla}^2\phi(\mathbf{x}, t) + m^2\phi(\mathbf{x}, t) \\
\implies \frac{\partial^2\phi(\mathbf{x}, t)}{\partial t^2} - \boldsymbol{\nabla}^2\phi(\mathbf{x}, t) &= -m^2\phi(\mathbf{x}, t) \\
\implies \left(\frac{\partial^2}{\partial t^2} - \boldsymbol{\nabla}^2 + m^2\right)\phi(\mathbf{x}, t) = 0 &
\end{aligned}$$

The above equation is called the *Klein-Gordon equation.* In covariant notation, we can compactly write it as:

$$\left(\partial_\mu\partial^\mu + m^2\right)\phi(x^\nu) = 0$$

The KG equation is Lorentz invariant, that is, under a Lorentz transformation, $x^\mu \to x'^\nu$, $\phi'(x'^\nu) = \phi(x^\mu)$

3.1.2 Continuity Equation

Taking the complex cojungate of the Schrödinger's equation and then after some algebraic manipulation, we obtain:

$$\frac{\partial(\psi^*\psi)}{\partial t} = \frac{i}{2m}\boldsymbol{\nabla}\cdot[\psi^*(\boldsymbol{\nabla}\psi) - \psi(\boldsymbol{\nabla}\psi^*)]$$

[1] A better analogy would be to use density matrices which gives the von Neumann equation which uses commutator bracket

Here, we identify:

$$\rho := \psi^*\psi \qquad\qquad \mathbf{J} = \frac{-i}{2m}[\psi^*(\boldsymbol{\nabla}\psi) - \psi(\boldsymbol{\nabla}\psi^*)]$$

which yields the well-known form of the *continuity equation*:

$$\frac{\partial \rho}{\partial t} + \boldsymbol{\nabla}\cdot\mathbf{J} = 0 \qquad \longrightarrow \qquad \partial_\mu J^\mu = 0$$

where $J^\mu = (\rho, \mathbf{J})$ is the four-current. Note that in this case, ρ is a positive-definite quantity and can indeed have the interpretation of probability. Also, note that if ψ somehow becomes real-valued, then $\mathbf{J} = 0$ which implies that ρ is constant in time (though it can change in space).

Doing the same thing to Klein-Gordon equation yields:

$$\frac{\partial}{\partial t}\left(\phi^*\frac{\partial \phi}{\partial t} - \phi\frac{\partial \phi^*}{\partial t}\right) - \boldsymbol{\nabla}\cdot(\phi^*\boldsymbol{\nabla}\phi - \phi\boldsymbol{\nabla}\phi^*) = 0$$

From this equation, we can identify:

$$\rho := \left(\phi^*\frac{\partial \phi}{\partial t} - \phi\frac{\partial \phi^*}{\partial t}\right) \qquad\qquad \mathbf{J} := -(\phi^*\boldsymbol{\nabla}\phi - \phi\boldsymbol{\nabla}\phi^*)$$

Thus here, the 4-current becomes $J^\mu = (\phi^*\partial^\mu\phi - \phi\partial^\mu\phi^*)$ Note the apparent problems with this identification:

- It is not apriori obvious that ρ is positive-definite and hence has problem with probability interpretation.
- The equation is second-order in time and hence ρ seems to have a term evolving forward and one term evolving backward in time.
- The dispersion relation does not have a single solution; the solutions are $\pm E$

The thing is, KG equation treats time and space on equal footing (unlike Schrödinger equation where time was in first order and space was in second order). Hence we have to consider all possibilities of moving back and forth in both space and time.

Lecture 04: Classical Fields

Let us start with a familiar setting, the spring-mass system in one-dimensions. So, each mass is connected with isotropic springs of natural length a and spring constant k, and at some instance, i^{th} spring is displaced from its equilibrium position by an amount η_i as shown below:

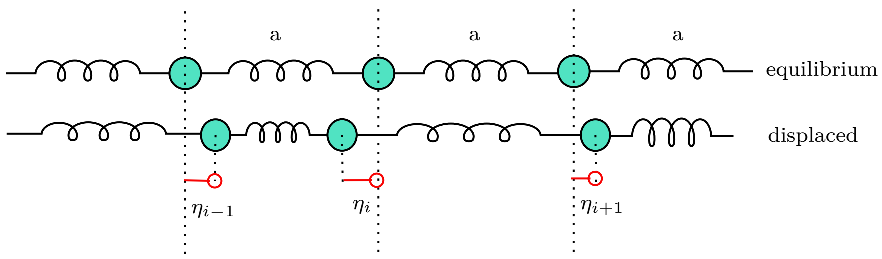


Figure 1: A spring-mass system; each mass is displaced by some amount from its equilibrium position

Then the kinetic and potential energies can be written as:

$$T = \frac{1}{2}\sum_i \dot{\eta}_i^2 \qquad\qquad V = \frac{1}{2}\sum_i k(\eta_{i+} - \eta_i)^2$$

From this, the Langrangian of the system is obtained as:

$$L = T - V = \frac{1}{2}\sum_i a\left[({}^m/_a)\dot{\eta}_i^2 - ka\left(\frac{\eta_{i+1}-\eta_i}{a}\right)^2\right] = a\sum_i L_i$$

The resulting ELEOM are:

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial L}{\partial \dot{\eta}_i}\right) = \frac{\partial L}{\partial \eta} \implies ({}^m/_a)\ddot{\eta}_i - ka\left(\frac{\eta_{i+1}-\eta_i}{a^2}\right) + ka\left(\frac{\eta_i - \eta_{i-1}}{a^2}\right) = 0$$

In the *continuum* limit, that is, $a \to 0$ (so, the system kind of becomes like a rod), we have:

$$\begin{aligned}
\frac{m}{a} &\longrightarrow \mu && : \text{mass per unit length of the rod}\\
ka &\longrightarrow Y && : \text{Young's modulus of the rod}\\
i &\longrightarrow x && : \text{position label}\\
\eta_i &\longrightarrow \eta(x) && : \text{field}\\
\frac{\eta_{i+1}-\eta_i}{a} &\longrightarrow \frac{d\eta}{dx} && : \text{derivative}\\
\sum_i &\longrightarrow \int dx && : \text{integral}
\end{aligned}$$

Hence in the continuum limit, the ELEOM becomes:

$$\lim_{a\to 0}\left(\mu\frac{\mathrm{d}^2\eta}{\mathrm{d}t^2} - \frac{Y}{a}\frac{\mathrm{d}\eta}{\mathrm{d}x}\bigg|_x + \frac{Y}{a}\frac{\mathrm{d}\eta}{\mathrm{d}x}\bigg|_{x-a}\right) = 0 \implies \mu\frac{\mathrm{d}^2\eta}{\mathrm{d}t^2} - Y\frac{\mathrm{d}^2\eta}{\mathrm{d}x^2} = 0$$

Defining the velocity as $v := \sqrt{\frac{Y}{\mu}}$, we obtain the wave-equation:

$$\frac{1}{v^2}\frac{\mathrm{d}^2\eta}{\mathrm{d}t^2} - \frac{\mathrm{d}^2\eta}{\mathrm{d}x^2} = 0$$

Note that, in this case, position becomes a mere label like in the discrete case, i just labelled the index of the i^{th} mass. Also, technically, η can also depend on time, so it is better to write it as $\eta(x,t)$.

Position and time, both are kind of labels and thus are independent. Hence, total and partial derivatives coincide in this case. Note that in continuum, the Langrangian itself becomes an integral of the form:

$$L = \frac{1}{2}\int\left[\mu\dot{\eta}^2 - Y\left(\frac{\mathrm{d}\eta}{\mathrm{d}x}\right)^2\right]dx$$

The bracketted term is the Lagrangian density, denoted by $\mathcal{L}$ and hence, the Lagrangian is the space integral of the Langrangian density:

$$L = \int \mathcal{L}\ dx$$

Henceforth, we allow ourselves to be *sloppy* enough and refer to Lagrangian density as simply the Langrangian when context is clear.

4.1 Action tells a lot!

Let us now consider a general setting. So, the Lagrangian density can be a function of the field, the spatial and temporal derivatives and space and time themselves. In compact notation, we write:

$$\mathcal{L} \equiv \mathcal{L}(\phi, \partial_\mu \phi, x^\mu)$$

Remember, that we define the classical action S as:

$$S = \int_{t_1}^{t_2} dt\ L = \int_{t_1}^{t_2} dt \int d^3\mathbf{x}\ \mathcal{L}(\phi, \partial_\mu \phi, x^\mu) = \int_{\mathcal{R}\subset\mathbb{R}^4} d^4\mathbf{x}\ \mathcal{L}(\phi, \partial_\mu \phi, x^\mu)$$

The integration can indeed be over a region $\mathcal{R}$ which is a subset of $\mathbb{R}^4$ in general. Mostly, we take $\mathcal{R} = [t_1, t_2] \times \mathbb{R}^3$.

We now want to use Hamilton's principle, which states that, under fixed boundary conditions, the system takes a path in configuration space for which the action is stationary, that is,

$$\delta S = 0$$

Note that the action is a functional which takes a field ϕ as input and returns a real number[1]. Let us consider a variation in the field itself: $\phi \longrightarrow\ \phi' = \phi + \delta\phi$ such that $\delta\phi$ vanishes at the boundary of the integration domain. Then we have the following:

$$\delta S = S[\phi'] - S[\phi] = \int d^4\mathbf{x}\ \big(\mathcal{L}(\phi', \partial_\mu \phi', x^\mu) - \mathcal{L}(\phi, \partial_\mu \phi, x^\mu)\big)$$

Now, we try to find what the above quantity is:

$$\begin{aligned}\mathcal{L}(\phi', \partial_\mu \phi', x^\mu) &= \mathcal{L}(\phi + \delta\phi, \partial_\mu(\phi + \delta\phi), x^\mu) \\ &= \mathcal{L}(\phi, \partial_\mu \phi, x^\mu) + \frac{\partial \mathcal{L}}{\partial \phi}\delta\phi + \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)}\delta(\partial_\mu \phi)\end{aligned}$$

Also note that $\delta(\partial_\mu\phi) = \partial_\mu(\delta\phi)$. Hence, from the above expression, we obtain that:

$$\delta S = \int d^4\mathbf{x}\ \Big(\frac{\partial \mathcal{L}}{\partial \phi}\delta\phi + \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)}\partial_\mu(\delta\phi)\Big)$$

Using *integration by parts* on the second term, we obtain:

$$\int d^4\mathbf{x}\ \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)}\partial_\mu(\delta\phi) = -\int d^4\mathbf{x}\ \partial_\mu\Big(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)}\Big)\delta\phi + \Big(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)}\delta\phi\Big)\Big|_{\text{boundary}}$$

The variation in the field vanishes at the boundary by our assumption. Then we obtain, for the variation of the action:

$$\delta S = \int d^4\mathbf{x}\ \Big(\frac{\partial \mathcal{L}}{\partial \phi} - \partial_\mu\Big(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)}\Big)\Big)\delta\phi = 0$$

Since the variation in the field was arbitrary, we obtain the Euler-Lagrange's equations for the field as:

$$\frac{\partial \mathcal{L}}{\partial \phi} - \partial_\mu\Big(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)}\Big) = 0$$

[1] For functionals, a common notation for the input is to use square brackets, that is, if A is a linear functional taking function f as input, $A[f] \in \mathbb{R}$

In general, we want the action to be *Lorentz* invariant, in order to maintain the universality of physical laws. Since the volume element of Minkowski space $d^4\mathbf{x}$ is invariant under Lorentz transformations, this imply that $\mathcal{L}$ should also be constructed of Lorentz-invariant quantities.

The field ϕ (and all its powers) is a Lorentz *scalar* and is hence invariant. The derivative $\partial_\mu\phi$ is not invariant, however, $\partial^\mu\partial_\mu\phi$ is. Bilinears like $(\partial^\mu\phi)(\partial_\mu\phi)$ are also invariant. Hence, a simple Lagrangian should be constructed of these quantities. Let us consider the Lagrangian:

$$\mathcal{L} = \frac{1}{2}(\partial^\mu\phi)(\partial_\mu\phi) - \frac{1}{2}m^2\phi^2 = \frac{1}{2}\eta^{\mu\nu}(\partial_\nu\phi)(\partial_\mu\phi) - \frac{1}{2}m^2\phi^2$$

Then, from the Lagrange's equations,

$$\begin{aligned}
\frac{\partial\mathcal{L}}{\partial\phi} &= -m^2\phi \\
\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)} &= \frac{1}{2}\eta^{\mu\nu}(\partial_\nu\phi + \partial_\mu\phi\delta^\mu{}_\nu) \\
&= \frac{1}{2}\eta^{\mu\nu}(2\times\partial_\nu\phi) \\
&= \eta^{\mu\nu}\partial_\nu\phi \\
&= \partial^\mu\phi
\end{aligned}$$

Then we finally obtain:

$$-m^2\phi - \partial_\mu\partial^\mu\phi = 0 \implies (\partial_\mu\partial^\mu + m^2)\phi = 0$$

This is the KG equation that we had seen earlier. Thus, the above Lagrangian models the KG equation.

When we have multiple fields $\{\phi_i\}$, then for each field, we will obtain:

$$\frac{\partial\mathcal{L}}{\partial\phi_i} - \partial_\mu\left(\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi_i)}\right) = 0$$

NOTE: In terms of the total Lagrangian, the ELEOM for the field is:

$$\frac{\delta L}{\delta\phi} - \partial_t\left(\frac{\delta L}{\delta\dot{\phi}}\right) = 0$$

Lecture 05 & 06: Looking into Hamiltonians

Previously, we saw a ton of crap about the Lagrangians and how the equations of motions are obtained from the action being stationary. Now, we will look into the *Hamiltonian* formalism which will allow us to quantise the field.

In basic classical mechanics, we had seen that given a Lagrangian $L(q_i, \dot{q}_i, t)$, we obtained the Hamiltonian by the Legendre transformation as:

$$H(q_i, p_i, t) = p_i\dot{q}_i - L(q_i, \dot{q}_i, t)$$

where $p_i := \frac{\partial L}{\partial\dot{q}_i}$ was defined to be the generalised momenta, conjugate to the generalised coordinate q_i. Similarly, for classical fields, we can define:

$$\pi^i(x^\mu) := \frac{\partial\mathcal{L}}{\partial(\partial_0\phi_i)} = \frac{\partial\mathcal{L}}{\partial\dot{\phi}_i}$$

This is the *canonical momentum* corresponding to the field. From the ELEOM, we obtain $\dot{\pi}^i = \frac{\delta L}{\delta \phi}$ Using this, the *Hamiltonian density* can be defined as:

$$\mathcal{H}(\phi^a, \pi_a, \partial_i \phi_a, x^\mu) := \pi^a \dot{\phi}_a - \mathcal{L}(\phi_a, \partial_\mu \phi_a, x^\mu)$$

In the final expression on the LHS, we have to replace $\dot{\phi}_a$ in terms of π^a to get the Hamiltonian density (Replacing this will get rid of all the time derivative and hence, in the Hamiltonian, we used ∂_i, the spatial derivatives only). Using this then, we get the total Hamiltonian as:

$$H[\pi^a, \phi_a] = \int d^3\mathbf{x}\ \mathcal{H}(\phi_a, \pi_a, \partial_i \phi_a, x^\mu)$$

5.1 Functional Derivatives

Since action, Lagrangian, etc. are functionals, an apt way to describe their variation is through the *functional derivative*. These come under the domain of functional analysis (which is Math ☠️), however, a short introduction might be necessary.

Let us take a functional $\mathcal{F}[f]$ and consider its variation $\delta\mathcal{F}$ when $f \to f + \delta f$, where $\delta f = \epsilon\eta$. Here ϵ is a small parameter and η is a generic function, often called the *test function*. Now,

$$\mathcal{F}[f + \epsilon\eta] = \mathcal{F}[f] + \left.\frac{\mathrm{d}\mathcal{F}[f + \epsilon\eta]}{\mathrm{d}\epsilon}\right|_{\epsilon=0} \epsilon + \mathcal{O}(\epsilon^2)$$

We define the functional derivative as the first-order coefficient of the expansion, that is,

$$\frac{\delta\mathcal{F}}{\delta f} = \lim_{\epsilon\to 0} \frac{\mathcal{F}[f + \epsilon\eta] - \mathcal{F}[f]}{\epsilon}$$

Functional derivatives seem to follow similar rules as in ordinary differential calculus (viz. linearity, Liebnitz rule, chain rule) and hence, can be treated to be operationally same as the ordinary derivative. However, exceptions are always there.

The differential (variation) of a functional is given as:

$$\delta\mathcal{F}[\phi] := \int \frac{\delta\mathcal{F}[\phi]}{\delta\phi} \delta\phi$$

Now let us try to relate functional derivative with normal derivative. For that, let us suppose that the space is discretised into cells, each of volume ΔV^i and each cell is assigned the *average value* of ϕ within the volume, that is,

$$\phi_i(t) = \frac{1}{\Delta V^i} \int\limits_{\Delta V^i} d^3\mathbf{x}\ \phi(\mathbf{x}, t)$$

Then, L now depends on the discrete components, ϕ_i and $\dot{\phi}_i$ and the variation can be written as in terms of partial derivatives (since ϕ_i are now discrete):

$$\delta L = \sum_i \left(\frac{\partial L}{\partial \phi_i} \delta\phi_i + \frac{\partial L}{\partial \dot{\phi}_i} \delta\dot{\phi}_i \right) = \sum_i \frac{1}{\Delta V^i} \left(\frac{\partial L}{\partial \phi_i} \delta\phi_i + \frac{\partial L}{\partial \dot{\phi}_i} \delta\dot{\phi}_i \right) \Delta V^i$$

Now, consider the Lagrangian variation using the definition of functional derivative:

$$\delta L = \int d^3\mathbf{x} \left(\frac{\delta L}{\delta \phi} \delta\phi + \frac{\delta L}{\delta \dot{\phi}} \delta\dot{\phi} \right)$$

From the two above equations, we can make the following identifications:

$$\frac{\delta L}{\delta \phi} = \lim_{\Delta V^i \to 0} \frac{1}{\Delta V^i} \frac{\partial L}{\partial \phi_i}$$
$$\frac{\delta L}{\delta \dot{\phi}} = \lim_{\Delta V^i \to 0} \frac{1}{\Delta V^i} \frac{\partial L}{\partial \dot{\phi}_i}$$

In the LHS, while taking functional derivative, $\mathbf{x}$ belongs to the i^{th} cell. Thus, we see that the functional derivative is proportional to the partial derivative. Now, given a Lagrangian, we can write the variation under $\phi \to \phi + \delta\phi$ using the Lagrangian density:

$$\delta L = \int d^3\mathbf{x}\ \frac{\partial \mathcal{L}}{\partial \phi}\delta\phi + \frac{\partial \mathcal{L}}{\partial(\partial_i \phi)}\delta(\partial_i \phi) + \frac{\partial \mathcal{L}}{\partial \dot{\phi}}\delta\dot{\phi}$$

Integration by parts and ignoring boundaries on the second term, we obtain:

$$\delta L = \int d^3\mathbf{x}\ \left(\frac{\partial \mathcal{L}}{\partial \phi} - \partial_i\Big(\frac{\partial \mathcal{L}}{\partial(\partial_i \phi)}\Big)\right)\delta\phi + \frac{\partial \mathcal{L}}{\partial \dot{\phi}}\delta\dot{\phi}$$

Comparing this with the definition of functional derivative, we obtain:

$$\frac{\delta L}{\delta \phi} = \frac{\partial \mathcal{L}}{\partial \phi} - \partial_i\Big(\frac{\partial \mathcal{L}}{\partial(\partial_i \phi)}\Big) \qquad \frac{\delta L}{\delta \dot{\phi}} = \frac{\partial \mathcal{L}}{\partial \dot{\phi}}$$

5.2 Hamilton's Equations of Motion

To obtain HEOM, we start with the variation of the total Hamiltonian,

$$\delta H = \int d^3\mathbf{x}\ \delta\big(\pi^\alpha \dot{\phi}_\alpha - \mathcal{L}\big) = \int d^3\mathbf{x}\ \big(\pi^\alpha \delta\dot{\phi}_\alpha + \dot{\phi}_\alpha \delta\pi^\alpha\big) - \delta L$$

Now, from ELEOM:

$$\delta L = \int \frac{\delta L}{\delta \phi_\alpha}\delta\phi_\alpha + \frac{\delta L}{\delta \dot{\phi}_\alpha}\delta\dot{\phi}_\alpha = \int \partial_t\Big(\frac{\delta L}{\delta \dot{\phi}_\alpha}\Big)\delta\phi_\alpha + \pi^\alpha \delta\dot{\phi}_\alpha = \int \dot{\pi}^\alpha \delta\phi_\alpha + \pi^\alpha \delta\dot{\phi}_\alpha$$

Using this, the variation in Hamiltonian becomes:

$$\delta H = \int d^3\mathbf{x}\ (\dot{\phi}_\alpha \delta\pi^\alpha - \dot{\pi}^\alpha \delta\phi_\alpha)$$

Thus, from above, we can identity:

$$\frac{\delta H}{\delta \phi_\alpha} = -\dot{\pi}^\alpha \qquad \frac{\delta H}{\delta \pi^\alpha} = \dot{\phi}_\alpha$$

Also, previously we saw the relation between functional derivative of the Lagrangian and partial derivative of the Lagrangian density. That could well be extended to the Hamiltonian and we obtain:

$$\dot{\pi}^\alpha = -\frac{\delta H}{\delta \phi_\alpha} = -\frac{\partial \mathcal{H}}{\partial \phi_\alpha} + \partial_i\Big(\frac{\partial \mathcal{H}}{\partial(\partial_i \phi_\alpha)}\Big) \qquad \dot{\phi}_\alpha = \frac{\delta H}{\delta \pi^\alpha} = \frac{\partial \mathcal{H}}{\partial \pi^\alpha} - \partial_i\Big(\frac{\partial \mathcal{H}}{\partial(\partial_i \pi^\alpha)}\Big)$$

In general, the Hamiltonian density does not depend on $\partial_i \pi^\alpha$, so we have $\dot{\phi}_\alpha = \frac{\partial \mathcal{H}}{\partial \pi^\alpha}$

5.3 Poisson Bracket

In the Hamiltonian formulation of classical mechanics, we defined the Poisson bracket of two quantities A, B by:

$$\{A, B\} = \frac{\partial A}{\partial q_i}\frac{\partial B}{\partial p_i} - \frac{\partial A}{\partial p_i}\frac{\partial B}{\partial q_i}$$

Similarly, for fields also, given two functionals $\mathcal{F}[\phi, \pi]$ and $\mathcal{G}[\phi, \pi]$, we define the Poisson bracket by:

$$\{\mathcal{F}, \mathcal{G}\} := \int d^3\mathbf{x}\ \frac{\partial \mathcal{F}}{\partial \phi_\alpha}\frac{\partial \mathcal{G}}{\partial \pi^\alpha} - \frac{\partial \mathcal{F}}{\partial \pi^\alpha}\frac{\partial \mathcal{G}}{\partial \phi_\alpha}$$

Given a function $\mathcal{F}[\phi^\alpha, \pi^\alpha, t]$, note:

$$\begin{aligned}
\dot{\mathcal{F}} &= \partial_t\mathcal{F} + \int d^3\mathbf{x}\left(\frac{\delta\mathcal{F}}{\delta\phi_\alpha}\frac{\partial\phi_\alpha}{\partial t} + \frac{\delta\mathcal{F}}{\delta\pi_\alpha}\frac{\partial\pi_\alpha}{\partial t}\right) \\
&= \partial_t\mathcal{F} + \int d^3\mathbf{x}\left(\frac{\delta\mathcal{F}}{\delta\phi_\alpha}\frac{\delta H}{\delta\pi^\alpha} - \frac{\delta\mathcal{F}}{\delta\pi_\alpha}\frac{\delta H}{\delta\phi_\alpha}\right) \qquad \text{(from HEOM)} \\
&= \partial_t\mathcal{F} + \{\mathcal{F}, H\}
\end{aligned}$$

This specifies the *time-evolution* of the functionals using the Poisson bracket. Now, let us consider the Poisson bracket of the fields ϕ_α and π^α. However, since the Poisson bracket was defined for functionals only, there is some problem. This can be resolved by noting that any function can be written as a functional depending on itself as:

$$\phi_\alpha(\mathbf{x}, t) = \int d^3\ \mathbf{x}'\phi_\alpha(\mathbf{x}', t)\delta^3(\mathbf{x} - \mathbf{x}')$$

From the definition of the functional derivative, it is easy to note that:

$$\frac{\delta\phi_\alpha(\mathbf{x}, t)}{\delta\phi_\beta(\mathbf{x}', t)} = \delta_{\alpha\beta}\delta^3(\mathbf{x} - \mathbf{x}')$$

Also, ϕ and π are *independent functionals* and hence,

$$\frac{\delta\pi(\mathbf{x}, t)}{\delta\phi(\mathbf{x}', t)} = 0 \qquad \frac{\delta\phi(\mathbf{x}, t)}{\delta\pi(\mathbf{x}', t)} = 0\ \forall\ \mathbf{x}, \mathbf{x}', t$$

Now, we can compute the Poisson bracket of the fields themselves and we obtain:

$$\begin{aligned}
\{\phi_\alpha(\mathbf{x}, t), \pi^\beta(\mathbf{x}', t)\} &= \int d^3\left(\mathbf{x}''\frac{\delta\phi_\alpha(\mathbf{x}, t)}{\delta\phi_\mu(\mathbf{x}'', t)}\ \frac{\delta\pi^\beta(\mathbf{x}', t)}{\delta\pi^\mu(\mathbf{x}'', t)} - 0\right) \\
&= \int d^3\left(\mathbf{x}''\delta_{\alpha\mu}\delta_{\beta\mu}\delta^3(\mathbf{x} - \mathbf{x}'')\delta^3(\mathbf{x}' - \mathbf{x}'')\right) \\
&= \delta_{\alpha\beta}\delta^3(\mathbf{x} - \mathbf{x}')
\end{aligned}$$

We also have the other two Poisson brackets which are trivial:

$$\{\phi_\alpha, \phi_\beta\} = 0 \qquad \{\pi^\alpha, \pi^\beta\} = 0$$

Lecture 07 & 08: Representations of Rotation and Lorentz group

A *group* is a tuple $(G, *)$ which follows certain rules like associativity, existence of identity, inverses, etc. A *Lie Group* [1] is some continuous group with properties so special that it becomes

[1] A Lie Group is a group that is also a manifold.

very important (and cool) to study them. [1].

I think a short, vague introduction cannot do justice to these elegant topics which are subjects on their own right and require proper analysis. Hence, I might skip them here (instead try to include them in the link below as and when I read about them)!

Lecture 09: Noether's Theorem

[1] A very basic and nasty introduction can be found here!